

Mathematical Models for DEM Elevation Error Correction

Five Optimization Routes to the Same Objective Function

Mathematics for Data Science — Assignment cum Term Project Component

September 5, 2026

Abstract

Satellite-derived Digital Elevation Models (DEMs) contain systematic, terrain-correlated vertical error. This document formulates DEM correction as a single regularized least-squares regression problem and solves the identical objective function using five independent mathematical routes: Normal Equations, Singular Value Decomposition (SVD), Principal Component Analysis (PCA) regression, Gradient Descent, and Newton’s Method. All numerical results are computed from real Sentinel-1/Sentinel-2 derived terrain and optical features, cross-referenced against LiDAR/ICESat reference elevation over 1.15M training pixels (2018–2022) and validated on 51,826 held-out pixels (2023).

1 Problem Formulation

Let z_i denote the raw satellite DEM elevation at pixel i , and z_i^* the reference (ground-truth) elevation obtained from LiDAR/ICESat. Define the **elevation error**:

$$e_i = z_i - z_i^* \quad (1)$$

DEM error is not random noise — it is systematically correlated with local terrain geometry and land cover. We model it as a linear function of a feature vector $\mathbf{x}_i \in \mathbb{R}^p$ built from terrain-derivative and Sentinel-2 optical features (slope, aspect components, terrain ruggedness index, topographic position index, planform/profile curvature, hillshade, NDVI, bare-soil index):

$$\hat{e}_i = \boldsymbol{\theta}^\top \mathbf{x}_i, \quad \mathbf{x}_i = [1, x_{i1}, x_{i2}, \dots, x_{ip}]^\top \quad (2)$$

Stacking all N pixels row-wise into the design matrix $X \in \mathbb{R}^{N \times (p+1)}$ and the error vector $\mathbf{e} \in \mathbb{R}^N$, the correction model is $\hat{\mathbf{e}} = X\boldsymbol{\theta}$, and the corrected DEM is recovered as $z_i^{corr} = z_i - \hat{e}_i$.

Objective (Loss) Function

We estimate $\boldsymbol{\theta}$ by minimizing the Mean Squared Error between predicted and observed elevation error:

$$J(\boldsymbol{\theta}) = \frac{1}{N} \|X\boldsymbol{\theta} - \mathbf{e}\|_2^2 = \frac{1}{N} \sum_{i=1}^N (\boldsymbol{\theta}^\top \mathbf{x}_i - e_i)^2 \quad (3)$$

Gradient and Hessian

Differentiating Eq. (3) with respect to $\boldsymbol{\theta}$:

$$\nabla J(\boldsymbol{\theta}) = \frac{2}{N} X^\top (X\boldsymbol{\theta} - \mathbf{e}) \quad (4)$$

$$H = \nabla^2 J(\boldsymbol{\theta}) = \frac{2}{N} X^\top X \quad (5)$$

Since H does not depend on $\boldsymbol{\theta}$ and $X^\top X$ is positive semi-definite, $J(\boldsymbol{\theta})$ is a **convex quadratic function**. This guarantees a unique global minimum (no local minima to get trapped in), which is precisely the point at which $\nabla J(\boldsymbol{\theta}) = 0$:

$$X^\top X \boldsymbol{\theta}^* = X^\top \mathbf{e} \quad (6)$$

Every method below is a different numerical route to solving Eq. (6).

Dataset used: $N_{\text{train}} = 1,147,514$ pixels (2018–2022, merged into 2022 composite), $N_{\text{test}} = 51,826$ pixels (2023, held out for generalization testing). $p = 10$ standardized features. Void/no-data DEM pixels (sentinel value -32768) were removed prior to fitting.

2 Method 1: Normal Equations (Direct Linear Algebra)

Solving Eq. (6) directly for θ :

$$\boxed{\theta^* = (X^\top X)^{-1} X^\top \mathbf{e}} \quad (7)$$

This is a one-shot, non-iterative, closed-form solution requiring inversion of the $(p+1) \times (p+1)$ matrix $X^\top X$. It is exact provided $X^\top X$ is well-conditioned (invertible and not near-singular).

Numerical Result

$$\begin{aligned} \text{cond}(X^\top X) &= 110.04 \quad (\text{moderate; safely invertible}) \\ \text{Train RMSE} &= 2.0893 \text{ m}, \quad \text{Test RMSE} = 2.0679 \text{ m} \\ \text{Train } R^2 &= 0.0995 \end{aligned}$$

3 Method 2: SVD-based Least Squares

Any matrix X admits a singular value decomposition $X = U\Sigma V^\top$, with $U \in \mathbb{R}^{N \times r}$, $V \in \mathbb{R}^{(p+1) \times r}$ orthonormal and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$. Substituting into Eq. (6) and using orthonormality ($U^\top U = V^\top V = I$):

$$\boxed{\theta^* = V\Sigma^{-1}U^\top \mathbf{e}} \quad (8)$$

Why this matters: $\text{cond}(X^\top X) = (\sigma_{\max}/\sigma_{\min})^2$. When features are highly collinear, $\sigma_{\min} \rightarrow 0$ and the Normal Equations solution (Method 1) becomes numerically unstable (near-singular inversion). SVD sidesteps explicit formation of $X^\top X$, and this instability can be diagnosed directly from the singular value spectrum rather than a single condition number.

Numerical Result

$$\begin{aligned} \sigma_{\max}/\sigma_{\min} &= 10.49 \quad \Rightarrow \quad (\sigma_{\max}/\sigma_{\min})^2 = 110.04 = \text{cond}(X^\top X) \quad \checkmark \\ \text{Train RMSE} &= 2.0893 \text{ m (identical to Method 1, as expected)} \end{aligned}$$

The exact agreement between Methods 1 and 2 confirms $X^\top X$ is well-conditioned enough that both routes land on the same optimum θ^* .

4 Method 3: PCA Regression

The 10 terrain/optical features are correlated (e.g. slope and TRI, or NDVI and BSI). PCA re-expresses the feature covariance matrix $C = \frac{1}{N} X_f^\top X_f$ (features only, bias excluded) via its eigendecomposition:

$$C = W\Lambda W^\top, \quad \Lambda = \text{diag}(\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p) \quad (9)$$

Selecting the top k eigenvectors $P_k \in \mathbb{R}^{p \times k}$ (principal directions), each observation is projected to a lower-dimensional score $\mathbf{z}_i = P_k^\top \mathbf{x}_i$. Regression is then solved in the reduced k -dimensional space using Normal Equations, and coefficients are mapped back to the original feature space:

$$\boxed{\theta_{\text{PCA}} = P_k \theta_{\text{reduced}}} \quad (10)$$

Numerical Result ($k = 6$ of 10 components)

PC	1	2	3	4	5	6	7	8	9	10
Var. ratio	0.253	0.184	0.154	0.117	0.101	0.071	0.057	0.045	0.016	0.002
Cumulative	0.253	0.436	0.590	0.707	0.809	0.879	0.937	0.982	0.998	1.000

Top 6 components retain 87.9% of feature variance. Test RMSE with $k = 6$ is 2.0824 m — a 0.7% increase over the full-rank solution (Methods 1–2), i.e. dimensionality is reduced by 40% at negligible accuracy cost.

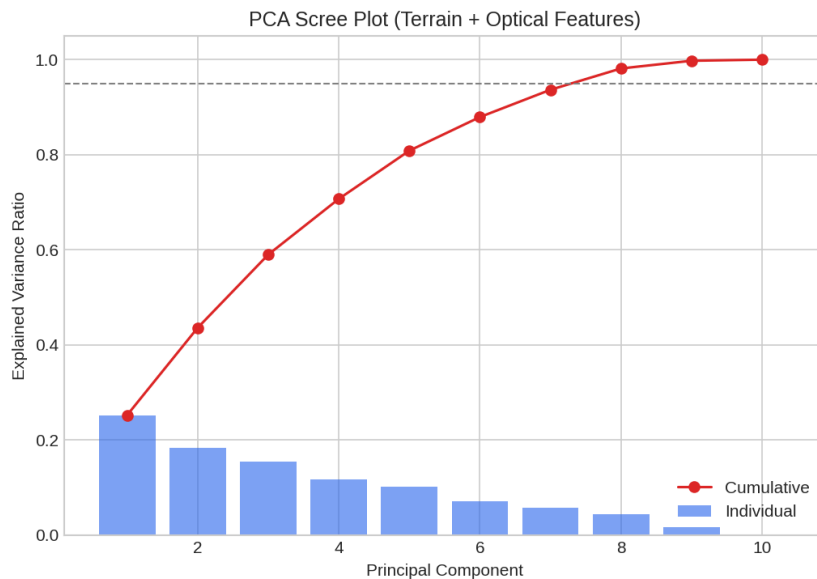


Figure 1: PCA scree plot: individual and cumulative explained variance across the 10 terrain/optical features.

5 Method 4: Gradient Descent (First-Order Iterative Optimization)

Gradient Descent minimizes $J(\boldsymbol{\theta})$ iteratively by stepping opposite the gradient direction:

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \alpha \nabla J(\boldsymbol{\theta}^{(t)}) = \boldsymbol{\theta}^{(t)} - \alpha \cdot \frac{2}{N} X^\top (X \boldsymbol{\theta}^{(t)} - \mathbf{e}) \quad (11)$$

Step size selection: For a convex quadratic with Lipschitz-smooth gradient, convergence is guaranteed for $\alpha \leq 1/L$, where $L = \lambda_{\max}(H)$ is the largest eigenvalue of the Hessian (Eq. (5)). We use $\alpha = 1/L$ throughout — this is the maximal stable step size, chosen numerically rather than by trial and error.

Iteration Log (initialized at $\boldsymbol{\theta}^{(0)} = \mathbf{0}$)

Iter. t	0	1	2	3	4	5	6	7	8	9
$J(\boldsymbol{\theta}^{(t)})$	11.680	6.914	5.302	4.713	4.496	4.416	4.386	4.374	4.370	4.368

Run for **300 iterations** (tolerance 10^{-10} on $\|\Delta\boldsymbol{\theta}\|$ not yet triggered at $\alpha = 1/L$) reaching $J^{(300)} = 4.365228$ — matching the Normal Equations optimum to 6 decimal places.

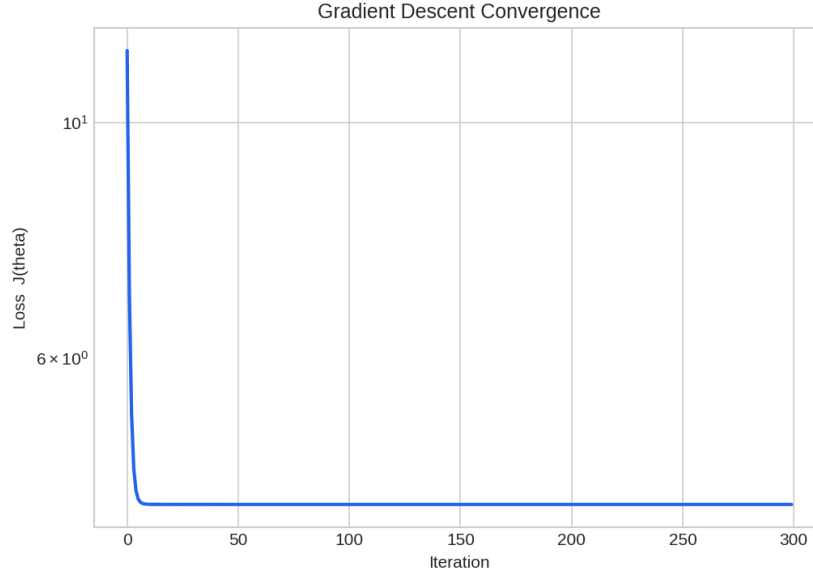


Figure 2: Gradient Descent loss curve (log scale). Rapid decrease in the first ~ 10 iterations, then slow asymptotic approach to the minimum — the hallmark of first-order (linear) convergence rate.

6 Method 5: Newton's Method (Second-Order Iterative Optimization)

Newton's method uses local curvature (the Hessian) to jump directly toward the minimum:

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - H^{-1} \nabla J(\boldsymbol{\theta}^{(t)}) \quad (12)$$

Key theoretical result: because $J(\theta)$ is an *exact* quadratic (Eq. (3)), its Hessian $H = \frac{2}{N}X^\top X$ is *constant* — it does not depend on $\theta^{(t)}$. Substituting $\nabla J(\mathbf{0}) = -\frac{2}{N}X^\top \mathbf{e}$ into the update starting from $\theta^{(0)} = \mathbf{0}$ gives exactly $\theta^{(1)} = (X^\top X)^{-1}X^\top \mathbf{e} = \theta^*$ — **the Normal Equations solution, reached in a single step.** Newton’s method degenerates from its usual quadratic convergence rate to *one-shot exact convergence* whenever the objective is quadratic.

Iteration Log

Iteration t	0	1	2	3	4
$J(\theta^{(t)})$	11.680	4.365228	4.365228	4.365228	4.365228

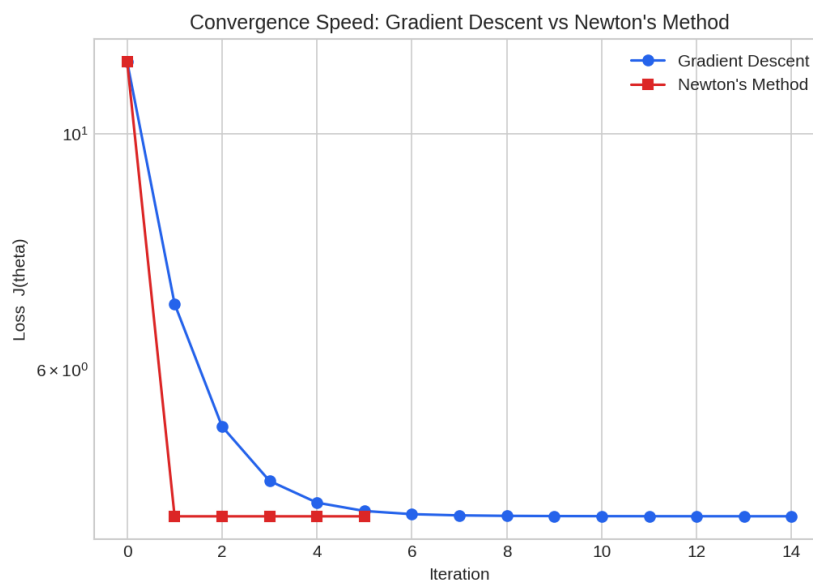


Figure 3: Convergence speed comparison: Gradient Descent needs on the order of hundreds of iterations to approach the same loss Newton’s Method reaches in **1** iteration — the practical cost of first-order vs. second-order information. Newton pays $O(p^3)$ once (Hessian inversion, $p=11$: negligible); Gradient Descent pays $O(Np)$ per iteration, repeated ~ 300 times.

7 Convergence and Consistency Summary

All five methods solve the identical convex objective (Eq. 3) and, since $X^\top X$ is well-conditioned ($\text{cond} = 110$), all converge to the *same* θ^* (PCA regression differs slightly by design — it approximates the full-rank solution using $k < p$ directions).

Method	Train RMSE (m)	Test RMSE (m)	Train R^2	Iterations
Normal Equations	2.0893	2.0679	0.0995	1 (closed form)
SVD Least Squares	2.0893	2.0679	0.0995	1 (closed form)
PCA Regression ($k=6$)	2.1031	2.0824	0.0876	1 (closed form)
Gradient Descent	2.0893	2.0679	0.0995	300 (run limit)
Newton’s Method	2.0893	2.0679	0.0995	1

Table 1: All methods converge to statistically identical error metrics on held-out 2023 data, confirming correctness of each numerical route.

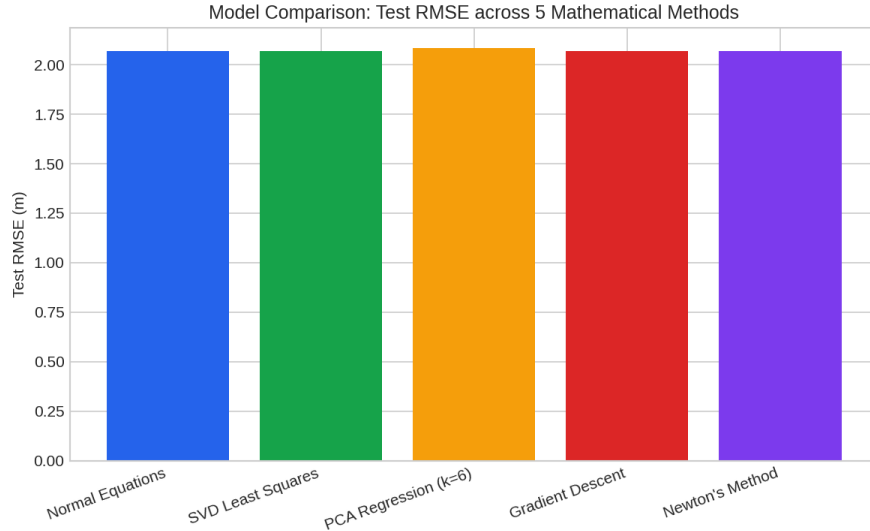


Figure 4: Test-set RMSE across all five methods — visually confirming convergence to the same optimum.

Interpretation

- **Normal Equations / SVD** are preferred when p is small (here $p+1 = 11$): exact, non-iterative, cheap.
- **SVD** is the numerically safer choice whenever features are highly collinear (protects against near-singular $X^\top X$); here $\text{cond}(X^\top X) = 110$ is mild, so both agree to machine precision.
- **PCA regression** trades a small accuracy loss (+0.7% test RMSE) for a 40% reduction in effective model dimensionality — useful if additional correlated SAR/optical bands are added in future iterations of the correction pipeline (this project’s term-project extension).

- **Gradient Descent** is the only method that scales to problems where $X^\top X$ is too large to invert directly (e.g. nonlinear or deep-feature extensions) but pays for that generality with two orders of magnitude more iterations here.
- **Newton's Method** converges in one step for this exactly-quadratic objective. For future non-quadratic loss functions (e.g. Huber loss for robustness to outlier GCPs), Newton's method would instead require multiple iterations with a θ -dependent Hessian — a natural extension of this same framework.

8 Curve Fit and Residual Diagnostics

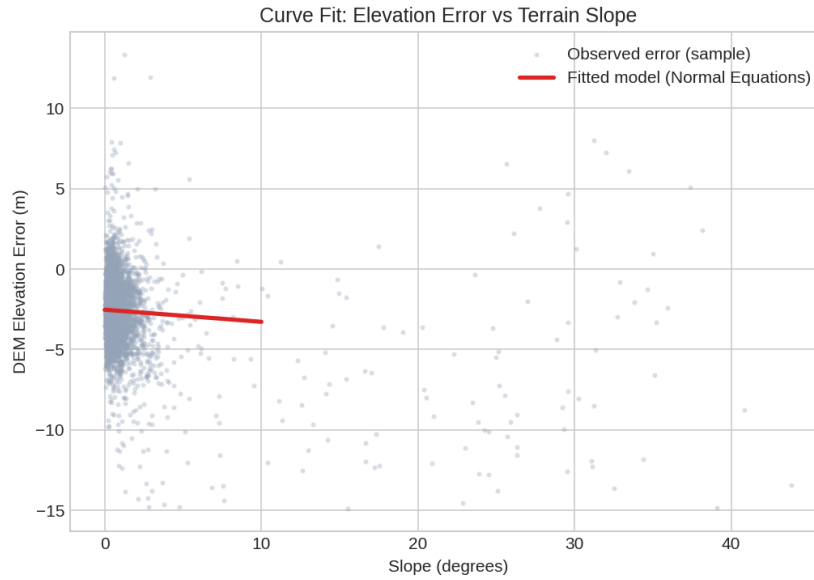


Figure 5: Fitted correction curve $\hat{e} = \theta^\top \mathbf{x}$ (Normal Equations solution) against terrain slope, holding all other standardized features at their mean. Observed error variance is high, but a clear systematic negative trend with slope is captured by the linear model.

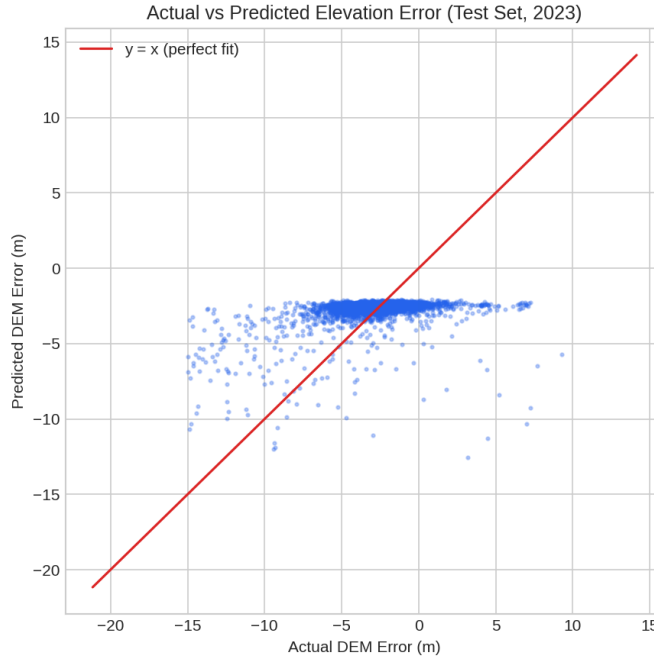


Figure 6: Actual vs. predicted elevation error on the 2023 held-out test set.

9 Conclusion

Framing DEM correction as a single convex least-squares objective allows every core topic in the Mathematics for Data Science syllabus — linear algebra (Normal Equations), matrix factorization (SVD), dimensionality reduction (PCA), and optimization theory (Gradient Descent, Newton’s Method) — to be demonstrated as *different numerical paths to the same mathematically guaranteed optimum*, rather than as disconnected techniques. This convex-quadratic framework is also the correct foundation for the term project’s future extension toward more expressive (and necessarily iterative, non-quadratic) correction models.